

D-F3 — Presentation script

Deliverable: Oral presentation script (10–15 minutes)

Sources: reports/Research_Report_Final.md · reports/Presentation_Outline_D-F2.md

Date: 29 July 2026

Total estimated duration: ~14 minutes 8 seconds (848 s) — adjustable by pacing results slides ± 30 s

Usage: Read naturally; [Figure: ...] marks where to advance to the slide visual.

Slide 1 — Title

Estimated time: 0:30

Good morning. Thank you for being here. My presentation is titled *Reproducible AI-Assisted Earthquake Early Warning Research for Western Seismic Regions*, with the subtitle *Method Development on a California FDSN Pilot with Transfer Path to European Networks*.

This talk summarises **AMGCR Earthquake Research** and the final research proposal and technical report—Deliverable D-F1—completed on **29 July 2026**, within the Swiss certificate programme in Assessment and Management of Geological and Climate Related Risk.

Everything I will describe comes from the completed pilot documented in that report and from repository artefacts frozen on **2026-07-29**. I am not presenting new computations today.

Transition: I will begin with why earthquake early warning and open methods matter for this programme, before clarifying how Europe and California fit together.

Slide 2 — Introduction: motivation

Estimated time: 0:35

Earthquake early warning aims to deliver alerts in the **seconds to tens of seconds** before destructive shaking arrives at a site. Operational systems rely on **P-wave detection, rapid magnitude estimation**, and **ground-motion prediction**.

In Europe and neighbouring regions—including the Mediterranean belt, the Alps, and areas with anthropogenic seismicity—this capability matters for civil protection. In Switzerland and more broadly in the EU, stakeholders increasingly expect **documented, open methodologies**, not opaque prototypes.

That expectation is the motivation for a certificate project that emphasises reproducible data handling, explicit processing steps, and honest limitation statements.

Transition: The same report makes clear that the programme’s geographic ambition and the pilot geography are not the same thing. That distinction is on the next slide.

Slide 3 — Programme positioning

Estimated time: 0:35

The **primary long-term target** of this research programme is **European seismic networks**. The **California IRIS and EarthScope pilot** is a **completed methods laboratory**—a place to validate workflow, scripts, and reporting—not the intended final deployment region.

The project began with an **EarthESND** reference implementation. That codebase is **complete** as a benchmark—the report cites **88 passing tests** on a full checkout—but it is **reference only** for this certificate narrative, not the scientific endpoint.

The completed active work uses **real waveforms**, theory, figures, tables, and formal reporting. All pilot science in this presentation reflects the freeze date **2026-07-29**.

Transition: Given that framing, it is worth stating briefly which gaps the workflow was designed to address.

Slide 4 — Research gap

Estimated time: 0:30

The report identifies four gap themes.

First, many state-of-the-art ML papers are demonstrated on region-specific archives, while **packaged, documented EEW workflows for European FDSN data** are less common at certificate level.

Second, studies often publish metrics without a full artifact trail from **raw MiniSEED** to **ML-ready feature matrices**.

Third, teams can conflate **implementing benchmark code** with completing **region-relevant science**; this project **decouples** EarthESND software from the Western pilot analysis.

Fourth, a small-**N** pilot can illustrate methodology but **cannot** support operational EEW claims; that distance remains explicit.

Transition: The objectives slide maps what was actually completed versus what remains proposed.

Slide 5 — Objectives

Estimated time: 0:40

The **primary objective** is a **reproducible, AI-ready EEW research workflow** aimed ultimately at **European networks**, documented for a Swiss and European research audience.

The **secondary objective**—the California pilot—is **completed**: acquisition through interpretation.

Objectives **O1 through O6** are **completed**: FDSN download, exploratory characterisation, onsite-style signal metrics and STA/LTA analysis, preprocessing toward ML inputs, an **8×18** feature matrix, and scientific interpretation including the European transfer narrative.

Objective **O4** is completed but with an important documented caveat: **instrument response removal was not applied—zero of eight traces** in the pilot run.

Objectives **O7 through O9**—European scale-up, AI training, and real-time prototyping—are **proposed**, not results of this pilot. Objective **O10**, the written proposal and report, is **completed**; D-F1 supersedes version one.

Transition: Before the dataset and pipeline, I will anchor the work in the literature the report cites.

Slide 6 — Literature: EEW fundamentals

Estimated time: 0:30

Classical onsite EEW uses triggers such as **STA/LTA** to estimate P-wave onset, then derives alert information from early windows of motion, as discussed by **Allen and Kanamori (2003)** and **Hoshiba et al. (2008)**.

Design trade-offs include **detection latency**, **false alarm rate**, and **magnitude saturation** for very large events.

European deployments additionally stress **standardised metadata**, **latency budgets**, and **uncertainty communication** to end users.

Our pilot signal phase uses ObsPy's **classic STA/LTA** with documented parameters—it is exploratory, not operational network tuning.

Transition: The data and software layer that implements this pilot sits alongside ML literature and reproducibility practice.

Slide 7 — Literature: data, software, and ML context

Estimated time: 0:35

ObsPy (Krischer et al., 2015) is the integration layer for MiniSEED, FDSN services, and processing in this repository.

For the California pilot, **USGS** supplies event metadata and **EarthScope** supplies waveforms—the same FDSN abstraction the report intends for **ORFEUS** and **EIDA** later.

Machine learning and reservoir approaches can learn from early waveform segments, but they require **large, diverse datasets** and careful **instrument response** handling.

EarthESND—multiscale ESN with dendritic readout—is cited in a **Japan K-NET** context. Here it exists as **reference pipeline code** in `src/`; the California pilot **did not** use those modules for the published results.

Reproducibility is institutionalised through **manifests**, **JSON configs**, and **phase reports**.

Transition: With that background, I turn to the concrete dataset the pilot analysed.

Slide 8 — California pilot dataset

Estimated time: 0:35

The completed pilot comprises **eight** earthquakes with magnitudes from **4.05 to 4.87**, mixing **mw** and **m1**, with origin times between **May and October 2024**.

Each retained event has one **BHZ** trace at **40 hertz**, in a **300-second** window starting at **catalog origin time**.

Stations in the manifest are **CI.ADO** for six events and **CI.USC** for two. The authoritative list is the CSV manifest under `data/manifests/`.

Exploratory inspection at one stage found **twelve** MiniSEED files on disk, including **four legacy** files outside the current manifest. **All completed analyses use the eight manifest-linked records only.**

No European waveforms are included in the completed pilot.

Transition: The next slide summarises how those eight records were selected from FDSN services.

Slide 9 — FDSN acquisition parameters

Estimated time: 0:30

Events were queried from **USGS** over a **2024** California bounding box with magnitude **4.0 to 7.5**, yielding **41** catalog matches before waveform filtering.

Waveforms were requested from **EarthScope** for candidate networks **CI, NC, and BK**, primarily **BHZ**, with **one successful trace per event**.

The pilot retained **eight** events with non-empty waveforms. Station choice follows a **fixed distance-ranked candidate list**, which is **not** uniform network sampling.

Access was **public FDSN**; **no credentials** were required. The download script is `scripts/download/download_iris_california_pilot.py`.

Transition: Having described the inputs, I will walk through the end-to-end workflow those scripts and analysis phases implement.

Slide 10 — End-to-end workflow

Estimated time: 0:38

The **completed** chain runs: **download**, then **Phase A.1 EDA**, **Phase A.2 signal analysis**, **Phase B.1 preprocessing**, **Phase B.2 feature engineering**, and **Phase C interpretation**.

Deliverables also include reproducibility and data availability statements—D-S1 and D-S2—and this **D-F1 final report**.

Outputs live under reports/—including `eda`, `signal_analysis`, `preprocessing`, `features`, and **figures**.

Critically, the **EarthESND modules in src/** **were not used** to produce these USA pilot results.

Proposed extensions—European FDSN acquisition, larger **N**, ML training, and **SeedLink**-style real-time work—are labelled as such in the report. Reproduction commands are in **Appendix B** of D-F1.

Transition: I will now split methods across two slides: first EDA and signal analysis, then preprocessing and features.

Slide 11 — Methodology: EDA and signal analysis

Estimated time: 0:32

Phase A.1 reads MiniSEED under `data/raw/iris/`, inspects files with ObsPy, and summarises **manifest-only** cohort statistics. It produces four exploratory figures.

Phase A.2 computes peak and RMS amplitudes, estimates noise from a **zero-to-five-second proxy window**, and reports peak SNR. It applies **STA/LTA** with **0.5-second STA**, **10-second LTA**, and threshold **2.5**, yielding a P pick and a signal-duration proxy.

Because downloads start at **origin time**, there is **no true pre-event quiet period**—a limitation that carries through to SNR interpretation.

Scripts: `run_california_pilot_eda.py` and `run_california_signal_analysis.py`.

Transition: Preprocessing and feature extraction build directly on those signal metrics and filtered-stage arrays.

Slide 12 — Methodology: preprocessing and features

Estimated time: 0:32

Phase B.1 applies demean, linear detrend, a **five-percent Hann taper**, and a **0.1 to 15 hertz** zero-phase band-pass. It attempts instrument response removal via EarthScope; in the completed run, response was **skipped for all eight traces**, so filtered data remain in **counts**. Stages are stored in NPZ files, with peak normalisation as the final step.

Phase B.2 extracts **eighteen features** from **filtered** traces, using P arrival times from Phase A.2. There are **nine** time-domain, **five** frequency-domain, and **four** earthquake-related features; FFT spectra are visualised but not stored as matrix columns.

PGM and Arias entries in the matrix are **counts-based proxies**, not physical PGA or standard Arias intensity.

Transition: With methods established, the next seven slides present results, beginning with exploratory views of the raw data.

Slide 13 — Results: EDA (example waveform)

Estimated time: 0:38

[Figure: `figures/example_waveform_raw.png` — Report Figure 1]

This panel shows the **highest magnitude** event in the manifest—**M 4.87** at **CI.ADO**—in **raw digital counts** over the full **300-second** window.

The cohort occupies a narrow **M 4 to 5** band, which reflects **query cutoffs**, not completeness of regional seismicity.

Strong motion is visible without correction, but the long window also shows baseline behaviour and coda; amplitudes are **not** ground motion in physical units.

This figure connects the abstract dataset table to what the signal and preprocessing stages actually receive.

Transition: The following slide summarises catalog-level views: magnitude distribution, timing, and station usage.

Slide 14 — Results: EDA (catalog views)

Estimated time: 0:38

[Figures: figures/magnitude_histogram.png, figures/events_over_time.png, figures/station_usage_frequency.png — Figures 2–4]

The magnitude histogram confirms the **small-sample** M 4–5 pilot band. The events-over-time plot shows clustering in **May, July through August**, and **late October 2024**; gaps reflect **download success**, not designed uniform sampling.

Station usage is heavily skewed: **ADO supplies six of eight** traces because of the acquisition script's distance-ranked list. That **structural bias** reappears in later feature comparisons.

Again, only **manifest-linked** files enter these summaries.

Transition: From catalogue statistics we move to waveform-level onsite metrics and STA/LTA behaviour.

Slide 15 — Results: signal analysis (high-SNR case)

Estimated time: 0:42

[Figure: figures/signal_analysis_ci40675215.png — Report Figure 5]

Across eight events, peak raw counts span roughly **1.7×10^4 to 3.5×10^6** . P picks range from **10.15 to 55.15 seconds** after trace start, with mean **31.3 seconds**. Peak SNR runs from **6.3 to 778**, median **49.9**.

Event **ci40675215**—magnitude **4.87**—is the favourable case: early pick near **10.2 seconds** and high SNR. The four-panel figure shows raw and normalised traces, the STA/LTA curve crossing threshold **2.5**, and the pick marker.

This is the kind of record where onsite triggering aligns intuitively with warning-time discussions—subject to the noise-window caveat.

Transition: Not every record behaves that way; the USC example on the next slide illustrates the limitation.

Slide 16 — Results: signal analysis (low-SNR / USC)

Estimated time: 0:38

[Figure: figures/signal_analysis_ci40699207.png — Report Figure 6]

At station level, **CI.ADO**—six events—shows mean peak SNR **169** and mean peak about **2.98×10^5 counts**. **CI.USC**—two events—shows mean SNR **42** but higher mean peaks, about **1.83×10^6 counts**.

Event **ci40699207** at USC has SNR near **six** and **zero seconds** signal duration by the post-P proxy—consistent with a **contaminated noise window** when the trace starts at origin.

The report treats SNR here as a **quality indicator**, not a calibrated detection probability.

Transition: Preprocessing next shows whether band-limited processing stabilises these traces for comparison and for feature extraction.

Slide 17 — Results: preprocessing

Estimated time: 0:38

[Figures: figures/preprocessing_ci40675215.png, figures/preprocessing_ci40699207.png — Figures 7–8]

Aggregate mean peak-SNR rises from **137** on raw traces to **1684** after filtering, before normalisation. Per-event filtered-to-raw ratios range **2.5 to 27.6**, mean **17.1**.

The four-panel layouts show **raw, detrended, filtered, and normalised** stages for the same two events we just discussed.

Visually, preprocessing **stabilises** traces for ML-oriented comparison. Scientifically, without response correction, filtered **counts** still cannot be read as velocity or acceleration for magnitude estimation.

Transition: The feature engineering results compress each trace into an eighteen-dimensional vector and summarise cohort behaviour.

Slide 18 — Results: feature engineering (spectra and correlations)

Estimated time: 0:38

[Figures: figures/feature_engineering_fft_spectra.png,
figures/feature_correlation_heatmap.png — Figures 9–10]

The primary artefact is an **8×18** matrix in reports/features/feature_matrix.csv. Cohort mean **crest factor** is approximately **11.4**; dominant frequency and spectral centroid cluster near **2.3 to 2.6 hertz**, consistent with the **0.1–15 hertz** band-pass.

The correlation heatmap shows **strong collinearity** among amplitude-derived features. The report highlights **crest factor**, **signal and spectral entropy**, and **zero-crossing rate** as partially complementary shape information.

At **N equals eight**, these views are **illustrative only**. The completed pilot includes **no ML training** and **no predictive performance metrics**.

Transition: Distributions and station boxplots close the results section before interpretation.

Slide 19 — Results: feature distributions and stations

Estimated time: 0:35

[Figures: figures/feature_distributions.png,
figures/feature_boxplots_by_station.png — Figures 11–12]

Per-feature histograms act as **pilot dashboards**, not density estimates. Amplitude-related features show **large spread**, driven by station gain and path effects in counts.

Boxplots compare **ADO** and **USC** for selected features; **USC** drives upper tails on peak, RMS, and PGM proxies, while counts are **unbalanced—six versus two events**.

PGM and Arias columns remain **counts proxies** only. Feature extraction is **complete**; model training is **proposed** under objective O8.

Transition: I will now step back from individual figures to discuss what the integrated pipeline achieves—and what it does not claim.

Slide 20 — Discussion

Estimated time: 0:35

The pilot demonstrates a **script-driven, documented path** from FDSN services to **tabular features** and **normalised NPZ stages**, matching certificate expectations for ObsPy, real data, graphs, and tables.

STA/LTA panels connect EEW theory to **P-time features**, but **late picks** and **origin-aligned noise** mean trigger parameters cannot be responsibly tuned on this dataset alone.

The feature matrix supports **exploratory baselines** in principle—linear models or random forests—but the report is explicit: **no ML training or hold-out evaluation** was performed.

EarthESND remains available for **future** Western-data comparison with a **deviation log** from Japan-centric paper metrics; it was **not validated** on California in completed work.

Transition: The report separates enumerated pilot limitations from formal threats to validity; I treat those on the next two slides.

Slide 21 — Limitations

Estimated time: 0:32

Key limitations include **eight events** and a **single BHZ component; origin-aligned windows** that bias noise and SNR; **zero of eight** successful response removals; **station selection bias** toward ADO with only **two** USC records; **mixed magnitude types**; and **unvalidated P picks**.

Results are **not comparable** to EarthESND **K-NET** paper metrics. Four **legacy MiniSEED** files may exist on disk outside the manifest—the report warns they must not enter summaries without reconciliation.

These points are preconditions for any European operational narrative.

Transition: Threats to validity pair each issue with mitigations planned—not yet executed.

Slide 22 — Threats to validity

Estimated time: 0:32

Construct validity: using counts where physical motion is required—mitigation: cache StationXML and remove response.

Internal validity: SNR inflation after filtering with a fixed early window—mitigation: pre-origin download windows.

External validity: California CI networks differ from European networks—mitigation: rerun the same scripts on an **EIDA/ORFEUS** pilot.

Statistical conclusion validity: N equals eight—mitigation: larger catalogues and bootstrap confidence intervals.

Selection bias from the first eight successful downloads and **confirmation bias** from a fixed STA/LTA threshold are documented with mitigations such as stratified sampling and threshold sweeps.

All mitigations are **proposed future work**.

Transition: That leads naturally to why a California pilot still supports a Europe-forward programme—and what transfer would look like.

Slide 23 — European relevance and transfer

Estimated time: 0:35

Europe is the primary programme focus; California validates **methods**, not final deployment geography.

The pilot demonstrates **ObsPy and FDSN reproducibility** and surfaces pitfalls—origin-aligned windows, response metadata, station bias—that also arise when integrating **ORFEUS and EIDA** data.

The **proposed transfer plan** reuses manifest structure, phase report templates, and the script layout with **European endpoint configuration**, documenting licensing, magnitude conventions, and channel choices per network.

No European waveform dataset is included yet. Expected challenges include fragmented access policies, lower event rates in some regions, heterogeneous station density, accelerometer versus broadband choices, and regulatory metadata requirements noted in the report.

Transition: I will close with planned extensions, conclusions, acknowledgements, and an invitation for questions.

Slide 24 — Future work, conclusion, acknowledgements, references

Estimated time: 0:45

Proposed future work includes expanding to **fifty to two hundred or more** events with stratified splits; a **European FDSN pilot**; fixing **instrument response** and adding **three-component** data and **pre-event noise windows**; **AI baselines** and EarthESND-inspired comparisons with a deviation log; **SeedLink** simulation for real-time latency; and formal publication of European transfer with uncertainty quantification.

Conclusions from the report: a **reproducible chain** to an **8×18 feature matrix** is **established**; raw waveforms show strong **station and path effects** and heterogeneous STA/LTA picks; preprocessing improves interpretability and SNR proxies but **response removal failed for all eight traces**, so **physical amplitude EEW is not yet supported**; features highlight redundancy among amplitudes and value in shape statistics at pilot scale; validity threats must be addressed before operational claims; **California proves the method—Europe is the intended scaled context**.

Acknowledgements: public data from **USGS** and **EarthScope** via FDSN, and the **ObsPy** project (Krischer et al., 2015). Key references include Allen and Kanamori (2003), Hoshiaba et al. (2008), and Joshi et al. (2026) for EarthESND as **reference only**. The full bibliography is in D-F1 section fourteen.

Thank you. I welcome your questions.

Timing summary

Slides	Block	Est. duration
1–5	Intro, gap, objectives	2:50
6–9	Literature and dataset	2:10
10–12	Workflow and methods	1:42
13–19	Results	4:29
20–24	Discussion, limits, Europe, close	2:57
Total	24 slides	~14:08

*To target **10 minutes**, shorten slides 13–19 by ~30%; to target **15 minutes**, expand results narration on Figures 5–12 by ~20%.*

Q&A preparation — ten likely examiner questions

Answers based solely on reports/Research_Report_Final.md.

1. Why use California data if the programme focus is Europe?

California provides a **completed FDSN methods laboratory** with mature USGS and EarthScope services and ObsPy practice. The report states Europe is the **primary long-term focus**; California validates **methods**, not deployment geography.

2. How many earthquakes and stations did you analyse?

Eight manifest-linked events, magnitudes **4.05–4.87**, each with one **BHZ** trace. Stations are **CI.ADO** (six events) and **CI.USC** (two events).

3. Did you train a machine learning model for magnitude prediction?

No. The completed pilot produces an **8×18 feature matrix** and normalized NPZ stages. The report states **no ML training or hold-out evaluation** and **no predictive performance metrics** in repository artefacts.

4. What is EarthESND's role in your results?

EarthESND is a **complete reference implementation (88 tests** on a full checkout) for **future comparative** studies. California pilot results **did not use** EarthESND src/ modules; K-NET scientific reproduction is **not started**.

5. Why were instrument responses not applied?

Preprocessing **attempted** response removal for all eight traces; **zero of eight** were applied in the completed run, and filtered traces remain in **counts**. The report documents this in preprocessing section 6.4 and limitations.

6. How reliable are your P-wave picks and SNR values?

P picks come from **STA/LTA** (0.5 s / 10 s, threshold **2.5**) and are **not analyst-validated**. Noise is estimated from the **first five seconds** of **origin-aligned** traces, so SNR is a **QC indicator**, not operational detection probability—especially for late picks and USC records such as **ci40699207**.

7. What does the preprocessing SNR increase from 137 to 1684 mean operationally?

The report interprets this as **processing sensitivity** with a **fixed early noise window**, **not** guaranteed operational EEW gain. Without response correction, filtered counts are still not physical ground motion.

8. Can you generalise feature correlations or station comparisons?

Not statistically. At **N=8**, distributions and correlations are **illustrative**. USC has **n=2**; ADO dominance reflects **acquisition policy**, not network-wide behaviour.

9. How would another researcher reproduce your work?

Appendix B of D-F1 and D-S1 document commands: download script, then four analysis scripts, from repository root with Python **≥3.11** and dependencies in `pyproject.toml`. Raw MiniSEED, logs, and PNG figures may be **gitignored** and must be **regenerated** or supplied separately per the audit and Appendix C.

10. What are the next concrete steps toward a European pilot?

Proposed: mirror the manifest and report pattern for **EIDA/ORFEUS**, configure European FDSN endpoints in the same script layout, document authentication and channel conventions, and rerun Phases A–B before any ML work—addressing response removal, pre-origin windows, and larger **N** as listed in future work (section 12.2).

Document control: Presentation script D-F3 v1.0 — AMGCR Earthquake Research, 29 July 2026. Derived only from D-F1 and D-F2. No source code modified.